I.21 · Shorten the route, enlarge the angle

If two straight lines from the endpoints of one side of a triangle meet inside it, the lines so joined will together be less than the remaining two sides of the triangle, but will contain a greater angle.

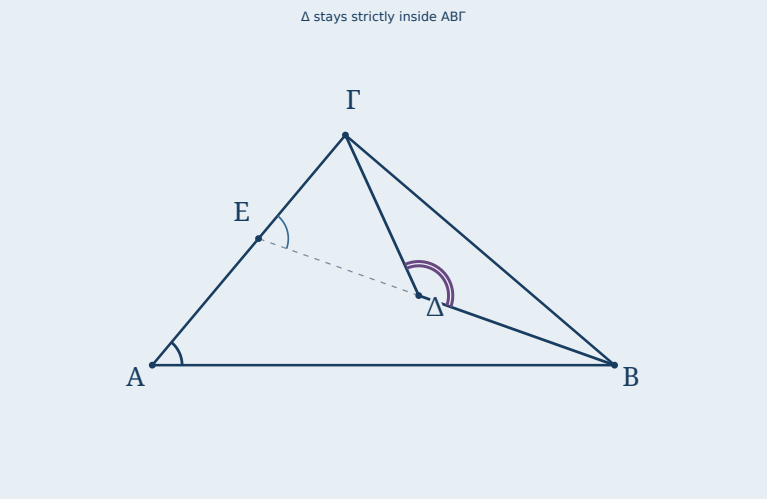


Plate at step 10: Keep both conclusions together.

Δ stays strictly inside ABΓ

E is 55% along AΓ and Δ is 55% along BE. These two position controls sample interior points while retaining the proof's hypotheses. The intermediate route B–E–Γ explains both inequalities. Drawing units and degree readings are modern illustrations, not premises of the proof.

Shorter route, larger angle

Compare	Outer · through A	Inner · through Δ
First leg · BA / BΔ	180.00	80.93
Second leg · AΓ / ΔΓ	117.00	68.72
Two-segment length	297.00	149.65
Included angle	50.00°	134.11°

BA + AΓ > BE + EΓ > BΔ + ΔΓ · ∠BΔΓ > ∠ΓEB > ∠BAΓ

The proposition

Ἐὰν τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν ἔσονται, μείζονα δὲ γωνίαν περιέξουσιν.	Si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent.	If two straight lines from the endpoints of one side of a triangle meet inside it, the lines so joined will together be less than the remaining two sides of the triangle, but will contain a greater angle.

1. Join the endpoints to an interior point Given

Τριγώνου γὰρ τοῦ ΑΒΓ ἐπὶ μιᾶς τῶν πλευρῶν τῆς ΒΓ ἀπὸ τῶν περάτων τῶν Β, Γ δύο εὐθεῖαι ἐντὸς συνεστάτωσαν αἱ ΒΔ, ΔΓ· λέγω, ὅτι αἱ ΒΔ, ΔΓ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τῶν ΒΑ, ΑΓ ἐλάσσονες μέν εἰσιν, μείζονα δὲ γωνίαν περιέχουσι τὴν ὑπὸ ΒΔΓ τῆς ὑπὸ ΒΑΓ.	In triangulo enim ΑΒΓ in uno latere ΒΓ a terminis Β, Γ duae rectae intus coniungantur ΒΔ, ΔΓ. dico, esse ΒΔ + ΔΓ < ΒΑ + ΑΓ et ∠ ΒΔΓ > ΒΑΓ.	In triangle ΑΒΓ, let ΒΔ and ΔΓ meet inside it from the endpoints Β and Γ of side ΒΓ. I say that ΒΔ and ΔΓ together are less than ΒΑ and ΑΓ together, but contain an angle ΒΔΓ greater than ΒΑΓ.

ἐντός / intus means inside: it is a hypothesis, not decoration. μέν … δέ balances the two surprising claims: smaller lengths, but a larger angle. Heiberg makes the sums explicit with + signs in Latin.

Reason: The task and its given objects A given object is part of the problem. It has not been constructed by this proof. A finite straight line here means a line segment.

Printed source: Greek pp. 50; Latin pp. 51.

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2. Continue ΒΔ to E on ΑΓ Construct

Διήχθω γὰρ ἡ ΒΔ ἐπὶ τὸ Ε.	educatur enim ΒΔ ad Ε.	For let ΒΔ be continued to E [on ΑΓ].

Διήχθω is a third-person perfect passive imperative: let it have been carried through. Latin educatur is a passive subjunctive used as an instruction. The English bracket explains the intersection supplied by the figure.

Reason: Postulate 2 · continue ΒΔ to the opposite side Because Δ is strictly inside ΑΒΓ, the ray from Β through Δ meets ΑΓ at an interior point E. The point ordering and intersection are geometric assumptions used by the source. A modern treatment must justify them; the drawing does not supply a separate proof.

Printed source: Greek pp. 50; Latin pp. 51.

3. Compare the first auxiliary triangle Prove

καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, τοῦ ΑΒΕ ἄρα τριγώνου αἱ δύο πλευραὶ αἱ ΑΒ, ΑΕ τῆς ΒΕ μείζονές εἰσιν·

et quoniam in quouis triangulo duo latera reliquo maiora sunt, in triangulo ABE erunt AB + AE > BE.

Since any two sides of a triangle together are greater than the remaining side, the sides AB and AE of triangle ABE are together greater than BE.

τῆς λοιπῆς is genitive after μείζονες: greater than the remaining side. The noun πλευρᾶς is understood. Latin reliquo is ablative of comparison, with latere understood.

Reason: I.20 · the two sides of ABE exceed the third Apply the triangle inequality in ABE: AB + AE > BE. These are three sides of one auxiliary triangle, not the two routes that the proposition ultimately compares.

Printed source: Greek pp. 50; Latin pp. 51.

4. Add the same remaining piece Prove

κοινὴ προσκείσθω ἡ ΕΓ· αἱ ἄρα ΒΑ, ΑΓ τῶν ΒΕ, ΕΓ μείζονές εἰσιν.

communis adiciatur ΕΓ. itaque ΒΑ + ΑΓ > ΒΕ + ΕΓ.

Let ΕΓ be added in common. Therefore BA and AΓ together are greater than BE and EΓ together.

κοινή / communis says the same segment is added to both sides of the inequality. AE and EΓ become the single whole AΓ. BA and AB name the same length.

Reason: Add the common segment ΕΓ Adding the same positive length preserves a strict inequality. Since E lies between A and Γ, AE + EΓ = AΓ. Thus the outer route B–A–Γ exceeds the intermediate route B–E–Γ. Heiberg cites expanded common notion 4 here; it is not the usual common notion 4 about coincidence.

Printed source: Greek pp. 50; Latin pp. 51.

5. Shorten the intermediate route once more Prove

πάλιν, ἐπεὶ τοῦ ΓΕΔ τριγώνου αἱ δύο πλευραὶ αἱ ΓΕ, ΕΔ τῆς ΓΔ μείζονές εἰσιν, κοινὴ προσκείσθω ἡ ΔΒ· αἱ ΓΕ, ΕΒ ἄρα τῶν ΓΔ, ΔΒ μείζονές εἰσιν.	rursus quoniam in ΓΕΔ triangulo ΓΕ + ΕΔ > ΓΔ, communis adiciatur ΔΒ. itaque ΓΕ + ΕΒ > ΓΔ + ΔΒ.	Again, since ΓΕ and ΕΔ in triangle ΓΕΔ are together greater than ΓΔ, let ΔΒ be added in common. Therefore ΓΕ and ΕΒ together are greater than ΓΔ and ΔΒ together.

πάλιν / rursus announces a second application of the same idea to a different triangle. ΕΔ plus ΔΒ becomes EB because the points occur in the order E, Δ, B.

Reason: I.20 again · use ΓΕΔ and add ΔΒ In ΓΕΔ, ΓΕ + ΕΔ > ΓΔ. Add ΔΒ to both sides. Because Δ lies between B and E, ΕΔ + ΔΒ = ΕΒ. Therefore the intermediate route Γ–E–B exceeds the inner route Γ–Δ–B. Its reversed reading direction changes no length.

Printed source: Greek pp. 50; Latin pp. 51.

6. Chain the length comparisons Prove

ἀλλὰ τῶν ΒΕ, ΕΓ μείζονες ἐδείχθησαν αἱ ΒΑ, ΑΓ· πολλῷ ἄρα αἱ ΒΑ, ΑΓ τῶν ΒΔ, ΔΓ μείζονές εἰσιν.	sed demonstratum est ΒΑ + ΑΓ > ΒΕ + ΕΓ. itaque multo magis ΒΑ + ΑΓ > ΒΔ + ΔΓ.	But ΒΑ and ΑΓ together were shown to exceed ΒΕ and ΕΓ together. Therefore, all the more, ΒΑ and ΑΓ together exceed ΒΔ and ΔΓ together.

ἐδείχθησαν is aorist passive: they were shown. πολλῷ and multo strengthen the comparison—by much, or all the more—without claiming a fixed numerical margin.

Reason: Chain the two strict length inequalities The outer route exceeds the intermediate route, which exceeds the inner route. Strict order is transitive. The total shortening equals the sum of the two positive triangle-inequality excesses; the measurements expose that identity as an additional modern observation.

Printed source: Greek pp. 50; Latin pp. 51.

7. Compare the exterior angle at Δ Prove

Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ ΓΔΕ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΓΕΔ.	rursus quoniam in quouis triangulo angulus extrinsecus positus maior est angulo interiore et opposito, in triangulo ΓΔΕ erit ∠ ΒΔΓ > ΓΕΔ.	Again, since an exterior angle of any triangle is greater than an interior and opposite angle, the exterior angle ΒΔΓ of triangle ΓΔΕ is greater than ΓΕΔ.

ἐκτός now means outside the small triangle ΓΔΕ, even though Δ is inside the original triangle. interiore et opposito preserves both qualifications: interior and opposite, not adjacent.

Reason: I.16 · the exterior angle at Δ In triangle ΓΔΕ, continue ΕΔ past Δ toward Β. The exterior angle ΒΔΓ exceeds the opposite interior angle ΓΕΔ. Since ΕΔ and ΕΒ are the same ray, ΓΕΔ equals ΓΕΒ. No theorem about the sum of three triangle angles is used.

Printed source: Greek pp. 52; Latin pp. 53.

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8. Compare the exterior angle at E Prove

διὰ ταὐτὰ τοίνυν καὶ τοῦ ΑΒΕ τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΕΒ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.	eadem de causa igitur etiam in triangulo ΑΒΕ erit ∠ ΓΕΒ > ΒΑΓ.	For the same reason, the exterior angle ΓΕΒ of triangle ΑΒΕ is also greater than ΒΑΓ.

διὰ ταὐτά means for the same reasons; ταὐτά contracts τὰ αὐτά. Latin eadem de causa uses the singular cause. The unchanged direction of rays ΕΔ and ΕΒ identifies the intermediate angle.

Reason: I.16 again · the exterior angle at E In triangle ΑΒΕ, continue ΑΕ past Ε toward Γ. Its exterior angle ΓΕΒ exceeds the opposite interior angle ΒΑΕ, which is ΒΑΓ because ΑΕ and ΑΓ are the same ray.

Printed source: Greek pp. 52; Latin pp. 53.

9. Chain the angle comparisons Prove

ἀλλὰ τῆς ὑπὸ ΓΕΒ μείζων ἐδείχθη ἡ ὑπὸ ΒΔΓ· πολλῷ ἄρα ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.

uerum demonstratum est ∠ ΒΔΓ > ΓΕΒ. multo igitur magis ΒΔΓ > ΒΑΓ.

But ΒΔΓ was shown to be greater than ΓΕΒ. Therefore, all the more, ΒΔΓ is greater than ΒΑΓ.

ἐδείχθη is singular here because its subject is one angle; the length argument used plural ἐδείχθησαν. The same rhetorical pattern carries a separate mathematical comparison.

Reason: Chain the angle comparisons The inner angle ΒΔΓ exceeds the intermediate angle ΓΕΒ, which exceeds the outer angle ΒΑΓ. This proves the angle claim independently of the length claim. A shorter pair of segments does not by itself prove a greater included angle.

Printed source: Greek pp. 52; Latin pp. 53.

10. Keep both conclusions together Conclude

Ἐὰν ἄρα τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν ἔσονται, μείζονα δὲ γωνίαν περιέξουσιν· ὅπερ ἔδει δεῖξαι.

Ergo si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent; quod erat demonstrandum.

Therefore, if two straight lines from the endpoints of one side of a triangle meet inside it, they will together be less than the remaining two sides, but contain a greater angle; which was to be proved.

συσταθῶσιν follows ἐάν as an aorist passive subjunctive. The futures ἔσονται and περιέξουσιν state what will follow. Latin minores erunt and maiorem … comprehendent reproduce the two consequences.

Reason: Both parts of the proposition are proved For the given interior meeting point, the inner two sides together are shorter, but their included angle is greater. The inequalities compare sums of lengths, not each corresponding side individually. The controls preserve the strict interior hypothesis.

Printed source: Greek pp. 52; Latin pp. 53.

One intermediate route, two different proofs

B–E–Γ sits between the outer and inner routes in length. The angle at E also sits between their included angles, in the reverse order. I.20 proves the length chain; I.16 proves the angle chain. Neither conclusion is inferred just by watching the controls. The numeric shortening is the sum of the two auxiliary triangles’ positive excesses.

Definitions and starting permissions used in this proof

Postulate 2 · β′ / II · Extend a straight line

Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ’ εὐθείας ἐκβαλεῖν.	Et ut recta linea terminata in directum educatur in continuum.	And that a finite straight line can be extended continuously in a straight line.

πεπερασμένην / terminata identifies a bounded segment. ἐπ’ εὐθείας / in directum keeps the extension straight: the command does not permit a turn at the endpoint.

Printed source: Greek pp. 8; Latin pp. 9.

Greek and Latin: Euclidis opera omnia, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1883, printed pp. 2–53. Public domain. University of Toronto scan. English, reading notes and redrawn coordinate diagrams: Stoicheia (2026), MIT. Greek ϑ is normalized to θ; line breaks and inline references are omitted. Latin u/v spelling is retained. Heiberg’s square brackets in I.1, I.4, I.8, I.13 and I.15 are retained. Printed angle and triangle symbols are represented as ∠ and △. The English is a reading translation, including explicit explanations of implied intersections.

The diagram is numerical evidence, not a formal proof. Length units and orientation are modern teaching choices.

The printed evidence

Source pages from Heiberg’s 1883 edition, including the passages and foundations used in this saved reading. Every image and the bundled font were checked against this edition before saving. The apparatus is available for inspection; it is not fully transcribed. Where the original scan is cropped, the page caption identifies that limit.

οὔτε ὀρθογώνιον· τὰ δὲ παρὰ ταῦτα τετράπλευρα τραπέζια καλείσθω.

κγ΄. Παράλληλοί εἰσιν εὐθεῖαι, αἵτινες ἐν τῷ αὐτῷ ἐπιπέδῳ οὖσαι καὶ ἐκβαλλόμεναι εἰς ἄπειρον ἐφ' ἑκάτερα τὰ μέρη ἐπὶ μηδέτερα συμπίπτουσιν ἀλλήλαις.

Αἰτήματα.

α΄. Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.

β΄. Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ' εὐθείας ἐκβαλεῖν.

γ΄. Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.

δ΄. Καὶ πάσας τὰς ὀρθὰς γωνίας ἴσας ἀλλήλαις εἶναι.

ε΄. Καὶ ἐὰν εἰς δύο εὐθείας εὐθεῖα ἐμπίπτουσα τὰς ἐντὸς καὶ ἐπὶ τὰ αὐτὰ μέρη γωνίας δύο ὀρθῶν ἐλάσσονας ποιῇ, ἐκβαλλομένας τὰς δύο εὐθείας ἐπ' ἄπειρον συμπίπτειν, ἐφ' ἃ μέρη εἰσὶν αἱ τῶν δύο ὀρθῶν ἐλάσσονες.

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23. Hero def. 71. Philoponus in anal. II fol. 18^{v}. Psellus p. 35. Martianus Capella VI, 712. Boetius p. 376, 23. αἰτ. 1—5. Martianus Capella VI, 722. Boetius p. 377, 4. Aspasius apud Simplicium in Arist. de coelo fol. 149: τὰ πέντε αἰτήματα. 1. Philop. in anal. II fol. 9^{v}. 10. 29. 2. Simplicius in phys. fol. 119. 3. Philop. in anal. II fol. 10. 29. 4. Id. ibid. fol. 10. 5. Id. ib. fol. 10. 29. Proclus p. 364, 14.

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1. τετράγωνα B. 2. τραπέζεια b. Def. 21 uulgo in 3, def. 22 in 5 diuidunt. 3. παράλληλοι δέ B. εὐθεῖαί εἰσιν Proclus, Psellus. 4. ἐς V. 5. συμπίπτειν P. ἀλλήλαις om. F. 6. αἰτήματα πέντε V, αἰτ. ἐστι πέντε BF, b m. 2. Numeros om. F. 9. ἐπ' εὐθείας κατὰ τὸ συνεχές PBFbp$_1$

liqua autem praeter haec quadrilatera trapezia adpellentur.

XXIII. Parallelae sunt lineae, quae in eodem plano positae et in utramque partem productae in infinitum in neutra parte concurrunt.

Postulata.

I. Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.

II. Et ut recta linea terminata in directum educatur in continuum.

III. Et ut quouis centro radioque circulus describatur.

IV. Et omnes rectos angulos inter se aequales esse.

V. Et, si in duas lineas rectas recta incidens angulos interiores et ad eandem partem duobus rectis minores effecerit, rectas illas in infinitum productas concurrere ad eandem partem, in qua sint anguli duobus rectis minores.

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receptum ordinem tuentur V, Proclus, Simplicius, Capella, Boetius, Campanus. 10. *ἐκβάλλειν* V. 11. *γράφεσθαι*] codd. omnes et Philoponus; *γράψαι* ex Proclo recepit August. 13. *ἀλλήλαις*] om. V. 15. *εὐθειά τις* P. 17. *ἐλάττονας* Proclus p. 191, 18 (non p. 364). *τὰς δύο*] PBVbp, *δύο* om. F, Proclus bis, Martianus Capella, Boetius, fort. recte. 18. *συμπίπτειν τὰς εὐθείας ἐκβαλλομένας ἐφ'* Proclus p. 364. *συμπίπτειν ἀλλήλαις* PV (*ἀλλήλαις* corr. ex *ἀλλήλας* P). 19. *ἐλάσσονες*] Pp, Proclus p. 364; *ἐλάττονες* uulgo. Dein add. *γωνίαι* FBVb, Philoponus; om. Proclus bis et Pp. In ed. Basil. et apud Gregorium *αἴτ.* 4—5 inter communes notiones (10—11) leguntur (*πᾶσαι αἱ ὀρθαὶ γωνίαι ἴσαι .. εἰσί; ἐκβαλλόμεναι αἱ .. εὐθεῖαι .. συμπεσοῦνται*). Post *αἴτ.* 5 in PF et V m. 2 et apud Campanum sequitur: *καὶ δύο εὐθείας χωρίον μὴ περιέχειν.*

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.

κα'.

Ἐὰν τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μὲν ἔσονται, μείζονα δὲ γωνίαν περιέξουσιν.

Τριγώνου γὰρ τοῦ ΑΒΓ ἐπὶ μιᾶς τῶν πλευρῶν τῆς ΒΓ ἀπὸ τῶν περάτων τῶν Β, Γ δύο εὐθεῖαι ἐντὸς συνεστάτωσαν αἱ ΒΔ, ΔΓ· λέγω, ὅτι αἱ ΒΔ, ΔΓ τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν τῶν ΒΑ, ΑΓ ἐλάσσονες μέν εἰσιν, μείζονα δὲ γωνίαν περιέχουσι τὴν ὑπὸ ΒΔΓ τῆς ὑπὸ ΒΑΓ.

Διήχθω γὰρ ἡ ΒΔ ἐπὶ τὸ Ε. καὶ ἐπεὶ παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσιν, τοῦ ΑΒΕ ἄρα τριγώνου αἱ δύο πλευραὶ αἱ ΑΒ, ΑΕ τῆς ΒΕ μείζονές εἰσιν· κοινὴ προσκείσθω ἡ ΕΓ· αἱ ἄρα ΒΑ, ΑΓ τῶν ΒΕ, ΕΓ μείζονές εἰσιν. πάλιν, ἐπεὶ τοῦ ΓΕΔ τριγώνου αἱ δύο πλευραὶ αἱ ΓΕ, ΕΔ τῆς ΓΔ μείζονές εἰσιν, κοινὴ προσκείσθω ἡ ΔΒ· αἱ ΓΕ, ΕΒ ἄρα τῶν ΓΔ, ΔΒ μείζονές εἰσιν. ἀλλὰ τῶν ΒΕ, ΕΓ μείζονες ἐδείχθησαν αἱ ΒΑ, ΑΓ· πολλῷ ἄρα αἱ ΒΑ, ΑΓ τῶν ΒΔ, ΔΓ μείζονές εἰσιν.

XXI. Schol. in Pappum III p. 1183, 4. Boetius p. 381, 26.

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2. εἰσιν P. 4. *πλευρῶν δύο εὐθεῖαι συσταθῶσιν ἐντὸς ἀπὸ τῶν περάτων ἀρξάμεναι αἱ* Proclus. 6. *δύο*] om. Proclus. 7. *ἐλάττους* F, Proclus. 8. *περιέξουσι* Proclus, Vbp. 11. *ΔΓ πλευραί τῶν* P. 13. *εἰσι* Vbp. *περιέχουσιν* PF.

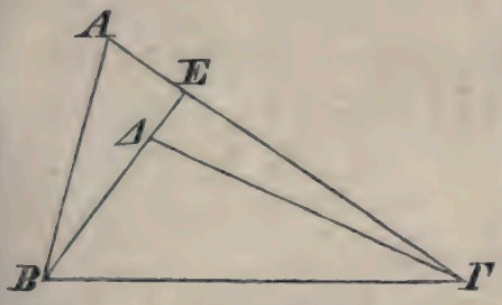
Ergo in quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta; quod erat demonstrandum.

XXI.

Si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent.

In triangulo enim $AB\Gamma$ in uno latere $B\Gamma$ a terminis B, Γ duae rectae intus coniungantur $B\Delta$, $\Delta\Gamma$. dico, esse $B\Delta + \Delta\Gamma < BA + A\Gamma$ et $\angle B\Delta\Gamma > BA\Gamma$.

educatur enim $B\Delta$ ad E. et quoniam in quouis triangulo duo latera reliquo maiora sunt [prop. XX], in triangulo ABE erunt $AB + AE > BE$. communis adiiciatur $E\Gamma$. itaque $BA + A\Gamma > BE + E\Gamma$ [*κ. ἔννν.* 4]. rursus quoniam in $\Gamma E\Delta$ triangulo



$$\Gamma E + E\Delta > \Gamma\Delta,$$

communis adiiciatur ΔB. itaque

$$\Gamma E + EB > \Gamma\Delta + \Delta B.$$

sed demonstratum est $BA + A\Gamma > BE + E\Gamma$. itaque multo magis $BA + A\Gamma > B\Delta + \Delta\Gamma$.

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14. $B\Delta\Gamma$] $\Gamma\Delta B$ F. 15. E] euan. F. 16. *εἰσιν*] PF; comp. b; *εἰσι* uulgo. 17. Post *πλευραί* in P del. *τῆς λοιπῆς μει.* 18. *εἰσιν*] PF; comp. b; *εἰσι* uulgo. *προσ-* supra m. 2 b. $E\Gamma$] $B\Gamma$ P. 19. *εἰσιν*] FP, comp. b; *εἰσι* uulgo. 20. $\Gamma E\Delta$] Δ add. m. 2 F. 21. *εἰσιν*] PFV; *εἰσι* uulgo. ΔB] $B\Delta$ b. 22. *ἄρα* ΓE, EB F. 23. BA] corr. in AB V. 24. $\Delta\Gamma$] $A\Gamma$ F. *εἰσιν*] PF; *εἰσι* uulgo.

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Πάλιν, ἐπεὶ παντὸς τριγώνου ἡ ἐκτὸς γωνία τῆς ἐντὸς καὶ ἀπεναντίον μείζων ἐστίν, τοῦ ΓΔΕ ἄρα τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΓΕΔ. διὰ ταὐτὰ τοίνυν καὶ τοῦ ΑΒΕ τριγώνου ἡ ἐκτὸς γωνία ἡ ὑπὸ ΓΕΒ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ. ἀλλὰ τῆς ὑπὸ ΓΕΒ μείζων ἐδείχθη ἡ ὑπὸ ΒΔΓ· πολλῷ ἄρα ἡ ὑπὸ ΒΔΓ μείζων ἐστὶ τῆς ὑπὸ ΒΑΓ.

Ἐὰν ἄρα τριγώνου ἐπὶ μιᾶς τῶν πλευρῶν ἀπὸ τῶν περάτων δύο εὐθεῖαι ἐντὸς συσταθῶσιν, αἱ συσταθεῖσαι τῶν λοιπῶν τοῦ τριγώνου δύο πλευρῶν ἐλάττονες μέν εἰσιν, μείζονα δὲ γωνίαν περιέχουσιν· ὅπερ ἔδει δεῖξαι.

κβ'.

Ἐκ τριῶν εὐθειῶν, αἵ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις [εὐθείαις], τρίγωνον συστήσασθαι· δεῖ δὲ τὰς δύο τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας [διὰ τὸ καὶ παντὸς τριγώνου τὰς δύο πλευρὰς τῆς λοιπῆς μείζονας εἶναι πάντῃ μεταλαμβανομένας].

Ἔστωσαν αἱ δοθεῖσαι τρεῖς εὐθεῖαι αἱ Α, Β, Γ, ὧν αἱ δύο τῆς λοιπῆς μείζονες ἔστωσαν πάντῃ μεταλαμβανόμεναι, αἱ μὲν Α, Β τῆς Γ, αἱ δὲ Α, Γ τῆς Β, καὶ ἔτι αἱ Β, Γ τῆς Α· δεῖ δὴ ἐκ τῶν ἴσων ταῖς Α, Β, Γ τρίγωνον συστήσασθαι.

Ἐκκείσθω τις εὐθεῖα ἡ ΔΕ πεπερασμένη μὲν κατὰ

XXII. Proclus p. 102, 16. Eutocius in Apollonium p. 10. Boetius p. 382, 1 (male). partem demonstrationis habet Proclus p. 330 sq.

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2. ἐντός] ἐν- in ras. b. ἐστίν] PF; ἐστί uulgo. ΓΔΕ] e corr. F m. 2; mutat. in ΓΕΔ V. ἄρα] supra F. 3.

rursus quoniam in quouis triangulo angulus extrinsecus positus maior est angulo interiore et opposito [prop. XVI], in triangulo $\Gamma\Delta E$ erit $\angle B\Delta\Gamma > \Gamma E\Delta$. eadem de causa igitur etiam in triangulo ABE erit $\angle\Gamma EB > BA\Gamma$. uerum demonstratum est $\angle B\Delta\Gamma > \Gamma EB$. multo igitur magis $B\Delta\Gamma > BA\Gamma$.

Ergo si in uno latere trianguli a terminis duae rectae intus coniunguntur, rectae coniunctae reliquis duobus lateribus trianguli minores erunt, maiorem autem angulum comprehendent; quod erat demonstrandum.

XXII.

Ex tribus rectis, quae tribus datis aequales sunt, triangulum construere (oportet autem duas reliqua maiores esse quoquo modo coniunctas [prop. XX]).

Sint tres datae rectae A, B, Γ, quarum duae reliqua maiores sint quoquo modo coniunctae, $A + B > \Gamma$, $A + \Gamma > B$, $B + \Gamma > A$. oportet igitur ex rectis aequalibus rectis A, B, Γ triangulum construere.

sumatur[1]) recta ΔE terminata in Δ, uersus E au-

1) Proclum non ipsa uerba Euclidis citare, adparet. cfr. idem p. 102, 19. Augustum perperam post $K\Lambda\Theta$ p. 54, 5. suppleuisse: *καὶ τεμνέτωσαν ἀλλήλους οἱ κύκλοι κατὰ τὸ K*, demonstraui „Studien“ p. 185.

$B\Delta\Gamma$] Δ in ras. F. *ἐστιν* PV. 4. $\Gamma E\Delta$] eras. F. *ταὐτά*] *τὰ αὐτά* F; *ταῦτα* Vbp. 5. *ἐστίν* P, ut lin. 7. 6. *ἀλλὰ καὶ τῆς* F. 7. $B\Delta\Gamma$] (alt.) $B\Delta$ in ras. sunt V. 12. *εἰσιν*] P; *εἰσι* uulgo. 15. *αἳ εἰσιν τρισὶ ταῖς δοθείσαις εὐθείαις ἴσαι* Proclus p. 329; sed p. 102: *αἳ εἰσιν ἴσαι τρισὶ ταῖς δοθείσαις εὐθείαις.* 16. *εὐθείαις*] om. b; m. rec. P; supra p; mg. m. 2 V; om. Eutocius. 17. *δέ*] Proclus, Eutocius; *δή* codd. *τάς*] corr. ex *ταῖς* F. *δύο*] *β* b. 18. *διὰ τὸ* — 20. *μεταλαμβανομένας*] omnes codd., Boetius; om. Proclus, Campanus; contra Eutocius ea habuisse uidetur. 21. *τρεῖς*] om. p.